

Roll No.....

Mid Semester Examination, September 2025

SET-I

Name of the Program: **B. Tech. Biotechnology**

Name of the Course: **Cell Biology**

Semester: **III**

Course Code: TBT302

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1		(10 marks)	
(a)	Summarize the major scientific theories regarding the origin of life, emphasizing the Miller-Urey experiment. How have these theories evolved with modern discoveries?		CO1
	OR		
(b)	Compare and contrast the classical cell theory with modern revisions. Discuss how exceptions to the cell theory are interpreted in contemporary cell biology.		CO1, CO2
Q2		(10 marks)	
(a)	Analyze the classification of cells within multicellular organisms, focusing on the differentiation between prokaryotic, eukaryotic, somatic, and germ cells, and discuss their roles in organismal function.		CO2
	OR		
(b)	Define the metabolic pool of a cell. Illustrate, with examples, how the major biochemical components (proteins, lipids, carbohydrates, nucleic acids) interact within metabolic pathways.		CO4, CO5
Q3		(10 marks)	
(a)	Discuss the structure and functions of the plasma membrane, comparing the fluid mosaic model with earlier membrane models. What evidence led to the acceptance of the fluid mosaic model as the standard?		CO3, CO4
	OR		
(b)	Discuss the various types of plastids found in plant cells, detailing their structural differences and biochemical functions.		CO3, CO4
Q4		(10 marks)	
(a)	Given a mixed cell population, explain two cell counting methods (e.g., hemocytometer counting and automated cell counters).		CO3, CO4
	OR		
(b)	Describe the mechanisms of solute transport across the plasma membrane, including passive and active transport.		CO3, CO4
Q5		(10 marks)	
(a)	Describe the main types of cell signaling pathways involved in cellular communication. How do cell surface receptors, particularly enzyme-linked receptors, mediate signal transduction, and what is their impact on cell behavior?		CO3, CO4
	OR		
(b)	What are chromosomal disorders? Provide a comprehensive explanation of the various types of chromosomal disorders, including numerical and structural abnormalities		CO3, CO4

